# RAHUL ARORA

**Research Scientist — Computer Graphics & Human-Computer Interaction**

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## EXECUTIVE SUMMARY

Research Scientist experienced in architecting multimodal AI wearables and end-to-end 3D graphics systems. Expert in owning sensing stacks (firmware to front-end) and driving ML research through ground-truth algorithms and user experiments. Widely published and patented in computer graphics, AR/VR, and HCI.

## CORE COMPETENCIES & TECHNICAL STACK

- **Domains:** HCI, Computer Graphics, AR/VR, Sensing, Gestural Interfaces.
- **Programming Languages:** Unity/C#, Python, C++, TypeScript/React, MATLAB.
- **Techniques:** Sensor Fusion, Spatial Tracking, Inertial Measurement Units (IMUs), Numerical Optimization, Geometry Processing, Quantitative/Qualitative User Studies, Rapid Prototyping.

## PROFESSIONAL EXPERIENCE

### ∞ META REALITY LABS — Dec 2021 – Jul 2026

*Research Scientist (Input Explorations Team)*

- **Meta Neural Band:** Led R&D for continuous-control gestures on Meta Ray-Ban Display, from sensor fusion and gesture models to user studies and OS integration for release. Contributed to a Nature (2025) paper and two patent applications [1, 2].
- **3D Math and Head Gestures:** Served as the primary resource for spatial math across Wearables, resolving on-device orientation-estimator issues, developing IMU-based head gestures, optimizing sensing stack for device performance, aligning coordinate conventions, and presenting technical guidance to partner teams.
- **Multimodal Interaction:** Developed gaze and hand interactions for Meta VR Glasses and CV/IMU-sensed hand tracking for Orion AR glasses; prototyped interaction models combining gaze and sEMG interactions.
- **Ground-Truth Systems:** Developed firmware/software for collecting ground truth interaction data; designed human-subject studies to measure and improve the performance of novel neuromotor algorithms.

### ADOBE RESEARCH — Summers 2019, 2018, 2013

*Research Intern (Multiple Teams)*

- **2019 — Stylized Fluid Animation:** Developed algorithms for creating stylized animations of 2D fluids from a few keyframes.
- **2018 — Gestural Animation in VR:** Researched and built novel gestural methods for authoring spatial animations directly within VR. Published at UIST 2019.
- **2013 — Computer Vision Data:** Built and publicly deployed a gamified crowdsourcing platform to accelerate and improve object recognition tasks for computer vision.

### AUTODESK RESEARCH — Winter 2017 & Summer 2016

*Research Intern (User Interface Research Team)*

- **2017 — SymbiosisSketch:** Developed an augmented reality tool for 3D concept sketching, mapping physical tablet surfaces to mid-air AR environments. Published at CHI 2018; granted a US patent.
- **2016 — Sketching Perception:** Conducted rigorous lab experiments and quantitative analyses to baseline human 3D sketching ability and perception limitations. Published at CHI 2017.

## SELECTED PUBLICATIONS

**A Generic Noninvasive Neuromotor Interface for Human–Computer Interaction** *(Nature 2025)*
The first high-bandwidth sEMG interface that works out-of-the-box for discrete selection, continuous navigation, and text-input via handwriting. Highly collaborative work across 245 contributors at Meta Reality Labs.

**Mid-Air Drawing of Curves on 3D Surfaces in Virtual Reality** *(SIGGRAPH/TOG 2021)*
An easy-to-implement algorithm for interactive drawing of curves on 3D surfaces. This open-sourced technique is based on the musculoskeletal limitations of mid-air input and was developed after analyzing ten novel alternatives.

**MagicalHands: Mid-Air Hand Gestures for Animating in VR** *(UIST 2019)*
An end-to-end system for authoring spatial animations in VR using mid-air hand gestures. The gesture-based controls were designed using an elicitation study with professional animators.

**Volumetric Michell Trusses for Parametric Design & Fabrication** *(SCF 2019)*
The first algorithm capable of designing volumetric Michell Trusses. This enables the creation of strong and lightweight objects which can be fabricated using additive manufacturing techniques and easily edited by manipulating an embedded parametrization.

**Experimental Evaluation of Sketching on Surfaces in VR** *(CHI 2017)*
A rigorous set of quantitative user studies to characterize the factors that affect the accuracy of mid-air sketching in VR. These foundational results have informed the design of many subsequent experiments and systems in the AR/VR research community (cited over 250 times each).

## EDUCATION

**UNIVERSITY OF TORONTO** **Sep 2015 – Jul 2021**
*Ph.D. in Computer Science; Advisor: Prof. Karan Singh, Dynamic Graphics Project Lab*

- **Focus:** Computer Graphics and HCI.
- **Thesis:** *Creative Visual Expression in Immersive 3D Environments* — Guidelines, algorithms, and tools for overcoming the perceptual and musculoskeletal limits of mid-air input for AR/VR creative workflows.

**INDIAN INSTITUTE OF TECHNOLOGY (IIT) KANPUR** **Jul 2010 – Jul 2015**
*Bachelor and Masters of Technology (B.Tech. & M.Tech.) in Computer Science and Engineering*

- **Masters Thesis:** *Exploring Design Space by Interpolating Between Multiple Sketches* — Developed a novel technique for interpolating between multiple early-stage design concepts.
- **Undergraduate Research:** Improved the computational complexity of the perfect matching problem for $K_{3,3}$-free or $K_5$-free graphs, an important class of non-planar graphs.

## SELECTED HONOURS & SERVICE

- **Research Awards:** CHI Best Paper Honorable Mention (2021), Adobe Research Fellowship (2019), Mitacs Accelerate Award (2016).
- **Academic Excellence:** IIT-K Academic Excellence Award (2013), IIT-K Merit Scholarship (2010–2014).
- **Invited Talks:** Invited technical talks and guest lectures at MIT, NYU, Université de Montréal, UBC Vancouver, and Autodesk Research.
- **Academic Service:** Frequent reviewer for top-tier graphics/HCI conferences including SIGGRAPH, SIGGRAPH Asia, UIST, and CHI. Program Committee Member for ICCV workshop on sketching, CHI late-breaking work, SIGGRAPH Asia XR program, and Pacific Graphics.